

MiniAgent - admin lightweight agent platform

MiniAgent is a lightweight agent platform (Chinese/English) for individuals and small teams. The backend is a FastAPI app that manages agents, knowledge bases, tools, LLM configs, auth/RBAC, and runtime features (RAG retrieval, SQL agent, web search). The frontend lives in management/ (vue-pure-admin).

High-level architecture

```
flowchart TB
    subgraph HTTP
        API["app/api/*"]
    end
    subgraph Core
        SC["ServiceContainer<br/>app/core/service_container.py"]
        CFG["Settings<br/>app/core/config.py"]
    end
    subgraph Runtime
        AF["AgentFactory"]
        AR["AgentRunner<br/>LangChain"]
        TB["ToolBuilder"]
    end
    subgraph KB
        DOC["KBDocumentService"]
        RET["KBRetrievalService"]
        SR["KBSmartRouterService"]
        VS["VectorStore + BM25"]
    end
    subgraph Data
        REPO["repositories/async_*"]
        SQLITE[(SQLite)]
        CHROMA[(ChromaDB)]
        DUCK[(DuckDB)]
    end

    API --> SC
    SC --> AF --> AR
    AF --> TB
    SC --> DOC & RET & SR
    RET --> VS
    REPO --> SQLITE
    VS --> CHROMA
    SC --> DUCK
```